

13.1 Chain-of-Thought

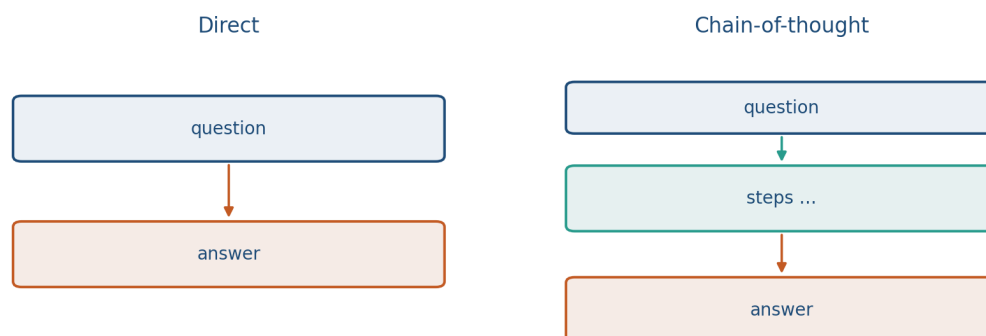
Ask for steps before the answer. Intermediate tokens can carry scratch work.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Chain-of-thought (CoT) is the habit of writing **steps** before the answer. You are not changing the weights. You are changing what the decoder is asked to emit.

1. Steps are part of the string

A direct prompt is “What is 17×24 ?” A CoT prompt is “Show your work, then give the number.” Few-shot CoT puts worked examples in the context; zero-shot CoT is often the sentence *Let’s think step by step*.



The model still predicts tokens. Intermediate tokens can allocate compute: they store partial products, units, or a plan. That helps **multi-step** arithmetic, symbolic puzzles, and some school-science items. It does not magically add retrieval (Week 14) or tools. A longer string is not “thinking” in the everyday sense; it is extra decoding you can inspect.

2. When it helps, when it does not

CoT tends to help when the gold answer needs several dependent hops. It can hurt on tasks that are a single lookup: extra steps are extra chances to wander. Longer traces cost latency and dollars. For a project, CoT is a **method knob**, not a personality.

You still need a metric on the **answer**, and a separate look at the trace (note **13.3**). A correct box after a wrong derivation is not the same system as a correct derivation. Self-consistency (note **13.2**) is what you do when one chain is too noisy to trust.

3. How to write it in a report

Say whether the prompt was zero-shot or few-shot, paste the exact instruction, and freeze decoding settings (temperature, max tokens). Do not claim the model “reasoned” unless you have a check on the steps. Lab 13 will grade traces as strings, with no API.

4. Practice

1. Why might CoT raise accuracy on a four-hop word problem and lower it on “What is the capital of France?”
2. Few-shot CoT puts examples in context. What goes wrong if those examples are a different task than the test item?
3. Name one number you would report besides final-answer accuracy if your project uses CoT.